

CONVICTION ENGINE v5

Complete System Reference — Internal Use

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1. The Simplest Explanation

1.1 The Problem This Engine Solves

The lemonade stand analogy:

Imagine you own a street full of lemonade stands. Each morning a thermometer (your quant model) tells you which stands are getting hotter — selling more lemonade, busier, trending up. But a rising temperature does not tell you if the lemonade stand is actually a great business or just having a good week. The Conviction Engine is the doctor's report on the business itself. It answers: is this a genuinely great business at a fair price?

The quant model generates price-based BUY/SELL signals. The Conviction Engine is a separate layer that sits on top of those signals and asks whether the underlying business justifies acting on them at today's price.

1.2 How the Engine Works — Five Steps

Step 1 — Is this even an equity?

First check: is this a real company with financial statements? A currency pair (EUR/USD), an ETF, or a commodity future has no earnings, no moat, no CEO. The engine immediately returns NOT_APPLICABLE for those. It only works on individual company stocks.

Step 2 — What kind of business is it?

The engine auto-detects the business type from financial data — no human input needed. Four types: SaaS (software subscriptions), compounder (high-quality growing business), income (pipelines, utilities, telecoms), cyclical (commodity/energy). A pipeline at 4× debt/earnings is normal and healthy. A tech startup at 4× debt/earnings is dangerous. Same number, completely different meaning. The engine uses different measuring sticks for each type.

Step 3 — Score the business quality (BQ score)

15 questions about the business itself — not the price. Is it growing? Does it make real cash? Is the CEO good? Does it have a moat? Is the debt manageable? Each yes adds points, each no subtracts. This produces the BQ raw score: a number from about -14 to +22. KXS (Kinaxis) scores +14. TELUS scores -5.

Step 4 — Apply the valuation tax

Even a perfect business is a bad buy if you pay too much. The valuation tax is always negative or zero — it only reduces the score. A SaaS company trading at 10× its annual revenue needs two things to go right simultaneously: the business has to keep executing AND investors have to keep paying that high multiple. That double requirement earns a heavy tax. A pipeline at 4× revenue is normal — no tax. BQ + valuation tax = conviction_score.

Step 5 — Apply the FS quality check and yield trap

The FS check asks: is this business viable as a going concern? Even if the conviction score looks acceptable, if the business is genuinely deteriorating it caps how much you can buy. The yield trap asks: is that big dividend actually a warning sign? If a company's dividend yield has spiked far above its own history AND is above the normal level for its stock market — the engine cancels any buy, regardless of score. These are hard gates. The final number is the conviction_score.

1.3 What the Engine Outputs

conviction_sc	What it means	BUY verdict	SELL verdict
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ore			
$\geq +8$	Outstanding business at a good price	MAX CONVICTION — full size	PAUSE SELL — keep 70%
$\geq +5$	Good business, some caution needed	TACTICAL BUY — 75% size	PARTIAL EXIT — keep 50%
$\geq +2$	Acceptable, proceed carefully	REDUCED BUY — 40% size	FULL EXIT — clean out
$< +2$	Do not buy / exit quickly	CANCEL BUY	HARD EXIT
Yield trap fires	Dividend is a danger signal	CANCEL BUY (hard gate)	HARD EXIT (hard gate)

1.4 Short-Term vs Long-Term Signals — The Two Position Layers

The quant model can fire signals at different time horizons. The engine tracks two separate layers for every holding:

Layer	What it is	Concrete example	Opens / Closes when
Core (long-term)	The structural thesis position — held for months to years	core_fraction = 0.75 → \$7,500 in KXS.TO on a long-interval BUY signal	Opens: long-interval BUY fires. Closes: long-interval SELL, or conviction drops below +2
Tactical (short-term)	A trading layer added ON TOP of the core for shorter momentum moves	tactical_fraction = 0.25 → \$2,500 added on a short-interval BUY. Total: \$10,000	Opens: short-interval BUY fires. Closes: short-interval SELL (core always untouched)

► Trailing-stop warning: if a short-interval BUY fires while the long-interval position is approaching its stop level — the engine flags this and cancels the tactical add. You do not add exposure just as the structural trade is about to exit.

1.3 Quant Model Functions — Registered in the Pipeline

The quant model that feeds signals into the conviction engine contains the following registered functions. Source file names are the internal Python filenames. The function keys (OSCILLATOR_DELTA etc.) are the branded names used throughout the codebase.

Function name (key)	Source file	Fixed interval or user-selectable	Signal logic (from source code)
OSCILLATOR_DELTA	divergence_com p.py	INTERVAL_DAILY (fixed)	Stochastic/price divergence. Tracks price vs indicator divergence at daily resolution. BUY: bullish divergence confirmed.

TRENDPULSE	trendline.py	All intervals D/W/M/Q/Y (user-selectable)	Trendline break signals. BUY/SELL when price breaks above/below computed trendlines.
BASELINEDIVERGENCE	general_divergence.py	INTERVAL_WEEKLY (fixed)	Price vs benchmark divergence on weekly candles. BUY: bullish divergence confirmed.
FRACTAL_TRACK	fib_ret.py	All intervals D/W/M/Q (selectable)	Fibonacci retracement tracking. Signals at key Fibonacci levels (23.6%, 38.2%, 50%, 61.8%, 78.6%).
BAND_MATRIX	bollinger.py	All intervals (user-selectable)	Bollinger Bands. BUY: Close crosses above middle band from below. SELL: crosses below from above.
DELTADRIFT	distance.py	All intervals (user-selectable)	MACD distance. BUY: MACD $\leq$ lower threshold AND EMA signal $\leq$ lower AND RSI $\leq$ 41 AND MACD bullish.
SIGMASHELL	sigma.py	INTERVAL_DAILY (fixed)	Statistical sigma. BUY: sigma < -3 AND price below 5-day ago price (mean-reversion signal).
PULSEGAUGE	sentiment.py	INTERVAL_WEEKLY (fixed for signals)	Sentiment/spread gauge. BUY: spread $>$ BUY_THRESHOLD AND RSI $<$ 50. SELL: spread $>$ SELL_THRESHOLD.
ALTITUDE_ALPHA	new_high.py	All intervals (user-selectable)	New highs breakout. BUY: price breaks above previous local maximum (altitude rising).
COMBINED_STRATEGY	combined_strategy.py	All intervals (user-selectable)	Composite: runs BAND_MATRIX + TRENDPULSE + DELTADRIFT together. Applies per-function success-rate filters.

Additional functions in codebase (not in main FUNCTION_CONFIG registry)

File	Fixed interval	What it does
fib_ext.py	INTERVAL_MONTHLY (fixed)	Fibonacci extension levels. Plots projected extension targets from major moves.
horizontal.py	All intervals (selectable)	Horizontal support/resistance levels. Identifies price levels where reversals cluster.
cycle_analysis.py	Various	Cycle trough detection. BUY at identified cycle troughs with higher-low and bullish phase conditions.
math_algo.py	D/W/M (selectable)	Statistical pair trading. Engle-Granger cointegration — Long/Short pairs when spread reverts.
fract_track_bands.py	INTERVAL_WEEKLY (fixed)	Fibonacci bands built from major upmoves ($\geq 20\%$). Used as input by combined_bands.py.
combined_bands.py	INTERVAL_WEEKLY (fixed)	Merged FTrack bands + FStack bands + horizontal weekly levels into unified support/resistance map.

- SHORT-TERM vs LONG-TERM SIGNALS — what this means in this engine:

When this document refers to a short-term or long-interval signal, it means a signal fired by a quant function running on a short interval (e.g. daily candles — signals play out over days to weeks) versus a long interval (e.g. monthly candles — signals play out over months). The `signal_timeframe` parameter is set by the quant function based on its own interval. The conviction engine does NOT determine the timeframe — it receives it as an input.

2. Master System Flowchart — End to End

Read top to bottom. Every signal goes through this exact sequence, in order.

STEP 0 — ASSET TYPE CHECK (first, before anything else)

Check asset quoteType → EQUITY: proceed to engine

ETF / INDEX / CURRENCY / CRYPTO / FUTURE → return NOT_APPLICABLE (no error thrown)

LAYER C — FULL QUARTERLY RECALCULATION (triggered, not scheduled)

Triggers: earnings date passed | manual override | buyback suspension | revenue miss | management change | dividend cut

1. Auto-detect business_type from sector/industry/payout data (manual override wins if set)
2. Pull financials: income statement, balance sheet, cashflow (one full data pull)
3. Compute 15 BQ dimensions — 7 auto from financials, 8 from manual overrides
4. Apply sector-calibrated scoring: Rule of 40 for saas/compounder; distribution coverage for income;
WACC: income 5.5%, compounder 7.5%, cyclical 9%, saas 10%; net debt thresholds vary by type
5. Compute valuation tax with sector-specific EV/rev tiers and floor triggers
6. Build PE history using contemporaneous trailing-4Q EPS (NOT today's EPS vs history)

LAYER A — PERSISTENT JSON STORE (conviction_store/{TICKER}.json)

STATIC (quarterly): bq_raw, bq_components, roic_5y_avg, pe_20y_array, eps_ttm, fcf_ttm, net_debt_stored, fwd_revenue_stored, annual_div_per_share_stored, fd_votes, fd_direction,

business_type, business_type_source, manual_overrides

DYNAMIC (daily): price, pe_ttm, pe_percentile_20y, ev_fwd_rev, oey, div_yield — updated every morning

LAYER B — DAILY LIGHTWEIGHT RUNNER (one price-only API call per ticker)

Fetch: price + market_cap (all other inputs read from stored JSON)

Recompute: pe_ttm | pe_pctile (vs 20Y array) | ev_fwd_rev | oey = fcf/mcap

div_yield | div_zscore vs 5Y history | valuation_tax | conviction_score = bq_raw + new_val_tax

fs_score = fs_quality_base + valuation slice (OEY, PE pctile, EV/rev) | fs_class | yield_trap

LAYER E — UNIVERSE RUNNER + DAILY REPORT GENERATOR

run_daily_universe(): calls daily_update() for all assets → alert_map {ticker:[flags]}

generate_daily_report(): universe health summary (strong/moderate/weak/conviction distribution)

On quant model BUY or SELL signal arriving:

HARD GATE: yield_trap = True?

BUY → CANCEL BUY | SELL → HARD EXIT (no score can override)

LAYER D — SIGNAL MODIFIER (reads stored data, <1 second, no API call)

1. Check signal_timeframe: 'long' (acts on core position) or 'short' (acts on tactical layer)

2. Apply FS graduated cap (Option B):

LONG-INTERVAL signals: fs=weak → conviction capped +1 | fs=moderate_low → conviction capped +4

SHORT-INTERVAL signals: fs=weak → conviction capped +2 | fs=moderate_low → no cap

3. Trailing-stop check: if short BUY + long position near stop → CANCEL tactical add

OUTPUT — SignalModification (verdict + sizing + position layers + rationale)

BUY: MAX CONVICTION (100%) | TACTICAL BUY (75%) | REDUCED BUY (40%) | CANCEL BUY (0%)

SELL: PAUSE SELL (retain 70% core) | PARTIAL EXIT (50%) | FULL EXIT | HARD EXIT
fd_direction adjusts sizing within tier | position_layers updated and persisted

3. Key Terminology — Every Term Defined

conviction_score

The single final number. BQ_RAW + valuation_tax (then FS cap applied). Range in practice: -12 to +10. One number, one decision per incoming signal.

BQ_RAW (Business Quality Raw Score)

Sum of 15 quality dimensions. No valuation. Answers: how good is this business intrinsically? Range: -14 to +22. KXS scores +14. TELUS scores -5.

valuation_tax

Always ≤ 0 . Applied to BQ to produce conviction_score. Sector-calibrated: SaaS floor at 4× EV/rev, income floor at 6×, compounder at 3×, cyclical has no floor.

OEY — Owner Earnings Yield

FCF / Market Cap. If you bought 100% of this business today, what annual cash yield would you earn? Buffett's primary valuation check. Sector-calibrated thresholds: saas $\geq 5\%$ = strong, compounder/cyclical $\geq 4\%$, income $\geq 3\%$. Below 1% = pricing in far-future earnings with no margin of safety.

fs_score (Fundamental Strength)

0–100 composite = fs_quality_base + daily valuation slice (OEY, PE pctile, EV/rev). Buckets into: strong / moderate_high / moderate / moderate_low / weak. Used to cap conviction_score before verdict mapping.

fs_quality_base

BQ_RAW rescaled: $50 + (BQ \times 2.5)$. BQ +10 → base 75. BQ -6 → base 35. Computed quarterly, static between earnings.

PE percentile (20Y history)

Where today's PE sits in 20 years of this stock's own PE history. Self-calibrating — no sector adjustment needed. 85th percentile = near bubble peak. 15th = crisis-level cheap. Built using contemporaneous trailing-4Q EPS.

fd_direction (Fundamental Direction)

5 direction votes: revenue / margins / FCF / EPS revision / buyback direction. Each positive/stable/negative. Majority wins. Positive fd_direction → bumps sizing up within tier (TACTICAL BUY: 75% → 85%). Negative → pulls sizing down (75% → 60%).

divergence_signal

Price $\geq 15\%$ below 52-week high for ≥ 60 days AND fd_direction $\neq$ negative. Auto-detected. The 60-day filter prevents fresh breakdowns from triggering falsely. Awards +2 to BQ score.

business_type

Auto-detected from sector/industry/payout data. Manual override wins if set. Four types: saas | compounder | income | cyclical. Drives: WACC rate, OEY threshold, net debt/EBITDA thresholds, EV/rev tax tiers, Rule of 40 applicability.

WACC (sector-calibrated)

The rate at which a business must earn on new investment to create value. income 5.5%, compounder 7.5%, cyclical 9%, saas 10%. Universal 8% WACC misprices regulated utilities.

distribution_coverage_ratio

For income businesses: FCF / annual dividend obligation. REPLACES Rule of 40. $>2.0\times$ = healthy. $\geq 1.2\times$ = sustainable. $<1.0\times$ = dividend is partly debt-funded = danger. TELUS at $0.87\times$ = correctly penalised.

yield_trap

Fires when BOTH: (1) dividend yield >1.5 std deviations above this stock's own 5Y mean, AND (2) yield exceeds market-absolute threshold (NZ: 12%, AU: 10%, CA: 7%, UK: 9%, US/India: 6%). Hard gate: CANCEL BUY or HARD EXIT, no override.

core_fraction / tactical_fraction

Position layer fractions. core_fraction = long-interval structural position (months to years). tactical_fraction = short-interval trading layer stacked on top. A short-interval SELL zeroes tactical_fraction only. Core stays unless long-interval signal exits.

ARR (Annual Recurring Revenue)

For SaaS only: annualised value of all active subscription contracts. Excludes one-time fees. ARR acceleration = the RATE of ARR growth is increasing (second derivative). $14\% \rightarrow 18\% \rightarrow 20\%$ is accelerating; $20\% \rightarrow 15\% \rightarrow 12\%$ is decelerating even though all positive.

10Y_Resilience

Binary label. "TOP 10% (Moat Intact)" if ROE $> 15\%$. Quick proxy check stored in record.

Margin of Safety

$= (1 - \text{PE_TTM} / 25) \times 100\%$. Percentage discount vs 25× PE premium baseline. Positive = trading below premium. Negative = above.

4. Scoring Architecture

4.1 Three-Stage Formula

Stage	Name	Range	Frequency	Core question
1	BQ_RAW	-14 to +22	Quarterly	How good is this business intrinsically?
2	valuation_tax	-10 to 0	Daily (price)	How much does entry price compress returns?
3	conviction_score = BQ + tax (then FS cap)	-12 to +10 (practical)	Daily	Do fundamentals support this trade at this price?

4.2 BQ Score — 15 Dimensions

Seven computed automatically from financial data. Eight require manual qualitative overrides. The margin_quality dimension is type-substituted per business type.

#	Dimension	Source	Range	What it measures
1	revenue_quality	Auto	-2 → +2	Gross margin, FCF conversion ≥90%, net retention ≥95%, RPO growth
2	growth_trajectory	Auto + manual	-2 → +2	Revenue growth level + is the RATE accelerating (2nd derivative)?
3	margin_quality (TYPE-SUBSTITUTED)	Auto	-2 → +2	SaaS/compounder: Rule of 40. Income: distribution coverage (FCF/annual_div). Cyclical: normalised FCF margin vs cycle.
4	balance_sheet	Auto	-2 → +2	Altman Z + net debt/EBITDA with sector-calibrated thresholds
5	roic_wacc_spread	Auto (5Y avg)	-1 → +2	5Y avg ROIC minus sector-calibrated WACC. Positive = value creation per \$ reinvested
6	gross_margin_trend	Auto	-1 → +1	Gross margin change over 3 years. Expanding ≥200bp = genuine pricing power
7	debt_maturity_risk	Auto	-1 → 0	Short-term debt >50% of total AND net leverage above cyclical safe threshold
8	ceo_quality	Manual 0-10	-1 → +2	Leadership quality + domain expertise. New CEO = additional -1 transition penalty
9	mgmt_capital_allocation	Manual 0-10	-1 → +2	SEPARATE from CEO. M&A track record, buyback timing, dividend discipline
10	competitive_moat	Manual 0-10	-1 → +2	Gartner rank, proprietary IP, switching costs, customer lock-in
11	macro_tailwind	Manual -1/0/1/	-1 → +2	Net macro environment for the business

		2		
1 2	divergence_signal	Auto-detected	0 or +2	Price $\geq 15\%$ below 52W high for ≥ 60 days AND fd_direction $\neq$ negative
1 3	deal_delay_risk	Manual bool	-1 $\rightarrow$ 0	Large enterprise deal timing slippage flagged in latest quarter
1 4	insider_ownership	Manual float	-1 $\rightarrow$ +2	>15% = founder-level alignment (+2). <1% = management treating it as a job (-1)
1 5	reinvestment runway	Manual float	-1 $\rightarrow$ +1	TAM / current revenue. <3 $\times$ = running out of room to compound at high ROIC

BALANCE SHEET SCORING — DEBT PURPOSE CLASSIFICATION

The engine classifies debt into three types before applying balance sheet penalties:

financial_engineering: net_cash > 0 (e.g. Apple, historically MSFT) $\rightarrow$ score +2 regardless of gross debt.

Rationale: debt issued deliberately at low rates to fund buybacks is not a credit risk.

capex_cycle: net_debt > 0 AND capex-to-revenue ratio increased >2 $\times$ in last 3 years (e.g. MSFT, GOOGL, META post-AI spend)

$\rightarrow$ score 0 or -1 with a flag. No full penalty unless short-term debt >50% of total.

Rationale: strategic long-dated investment debt with strong FCF coverage does not carry the same risk as operational leverage.

If an imminent declared capex expansion is found in earnings transcript $\rightarrow$ flag debt_purpose_forward: capex_cycle_expected $\rightarrow$ apply -1 anticipatory penalty.

operational: debt funding business operations (e.g. TELUS, pipelines) $\rightarrow$ full sector threshold penalties apply.

Rationale: operational leverage where FCF may not cover debt service is the primary credit risk scenario.

FCF is always a net concept (after capex, after interest, after tax) — "strong FCF" means strong on this net basis.

BQ_RAW = sum of all 15 dimensions. Max achievable $\approx$ +22 across ALL business types (type-substitution in dimension 3 ensures no type is disadvantaged on scale). Dims 1–7 auto-computed. Dims 8–15 require analyst manual override entry.

4.2b Fundamental Direction — 5 Votes and Thresholds

Fundamental Direction is computed from 5 independent votes. Majority wins (3 of 5). All votes are auto-computed from financial data — no manual input.

Vote	Positive trigger	Stable	Negative trigger	Source
Revenue	Revenue growth rate accelerating (rate of rate increasing) for 2+ consecutive quarters	Growth positive but flat	Growth decelerating 2+ quarters OR revenue declining	yfinance quarterly_income_st

				mt
Margins	Gross margin expanding ≥ 100 bp year-on-year	Within ± 100 bp	Gross margin compressing > 100 bp YoY	yfinance_income_stmt
FCF	FCF growing YoY AND FCF margin expanding	FCF positive but flat	FCF declining YoY OR turning negative	yfinance_cashflow
EPS revisions	Consensus EPS estimate for next 12M raised $\geq 3\%$ in past 90 days vs stored prior estimate	Within $\pm 3\%$	Estimate cut $\geq 3\%$ in 90 days	yfinance_eps_estimate_vs_stored_prior
Buybacks	$\geq 1\%$ of shares outstanding repurchased in trailing 12 months (shares_outstanding declined)	Programme active but $< 1\%$	Shares outstanding increased (dilution) OR buyback suspended	yfinance_shares_history_or_SEC_EDGAR

fd_direction = positive if ≥ 3 votes positive | negative if ≥ 3 votes negative | stable otherwise
 Sizing adjustment: positive $\rightarrow +10$ pp to base sizing % | negative $\rightarrow -15$ pp | stable $\rightarrow$ no change

Example: TACTICAL BUY base 75% + fd_positive $\rightarrow 85\%$. Same verdict, larger position.

EPS revisions: stored prior estimate from last monthly refresh is the comparison baseline.

Buybacks computed from: $(\text{shares_3y_ago} - \text{shares_current}) / \text{shares_3y_ago} \geq 0.01 =$ positive.

4.3 FS Score and Graduated Cap (Option B)

fs_quality_base = $50 + (\text{BQ_RAW} \times 2.5)$. Static between earnings. fs_score = fs_quality_base + daily valuation slice (OEY, PE pctile, EV/rev).

fs_score	fs_class	Cap on conviction — LONG-INTERVAL signals	Cap on conviction — SHORT-INTERVAL signals
≥ 75	strong	No cap	No cap
≥ 55	moderate_high	No cap	No cap
≥ 40	moderate	No cap	No cap
≥ 25	moderate_low	conviction = $\min(\text{score}, +4)$ — TACTICAL BUY max	No cap

<25	weak	conviction = min(score, +1) — REDUCED BUY max	conviction = min(score, +2)
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FS is applied as a POST-COMPUTATION cap on conviction_score. Flow: compute conviction → apply FS cap → apply yield trap hard gate → map to verdict. Short-interval signals get relaxed caps because a tactical trade does not require the same quality bar as a structural hold.

4b. Business Quality — Data Sources for All 15 Dimensions

Every dimension, its source, and how it is computed. No guesswork, no assumptions.

#	Dimension	Source	How computed / sourced
1	Revenue quality	yfinance	Gross margin from income_stmt. FCF conversion = FCF/net_income from cashflow. Net retention proxy from RPO growth if available.
2	Growth trajectory	yfinance	Revenue growth rate: quarterly_income_stmt. Revenue accelerating: last 3 quarters of growth rates, checks second derivative (last rate > prior rate > prior prior rate).
3	Margin quality	yfinance	Compounders/SaaS: Rule of 40 = revenue_growth% + FCF_margin%. Income type: distribution_coverage = FCF/annual_div_obligation. Cyclical: FCF margin vs prior year.
4	Balance sheet	yfinance + debt_purpose logic	Altman Z-score from balance_sheet + income_stmt. Net debt/EBITDA vs sector thresholds. Debt purpose classification applied first (see Section 4.2 note).
5	ROIC / WACC spread	yfinance (5Y avg)	ROIC = NOPAT / invested_capital from financials. WACC = sector rate (see Section 5.3). Spread = ROIC – WACC. 5-year average to smooth cycle effects.
6	Gross margin trend	yfinance	Gross margin 3 years ago vs today. Expansion ≥200bp = +1. Source: annual income_stmt gross_profit / revenue.
7	Debt maturity risk	yfinance	Short-term debt as % of total debt from balance_sheet. >50% AND net leverage above cyclical threshold = –1.
8	CEO quality	Claude API web search	Search: "[company] CEO [name] tenure track record domain expertise". Score 0–10 → maps –1/0/+1/+2. New CEO transition = additional –1 penalty.
9	Capital allocation	Claude API + SEC EDGAR	Buyback history from SEC EDGAR Form 4 + yfinance shares history. M&A track record from web search. Score 0–10.
10	Competitive moat	Claude API web search	Search: "[company] competitive moat switching costs customer lock-in". Adversarial prompt: explicitly asks for threats and disruption risk. Score 0–10.
11	Macro tailwind	Claude API web search	Search: "[company] macro environment headwinds tailwinds demand outlook [year]". Score –1/0/+1/+2. Confidence threshold: if agent confidence <0.7, default to 0.
12	Divergence signal	yfinance	current_price ≤ hist_52W_high × 0.85 (≥15% below) AND days_below_high ≥ 60 AND fd_direction ≠ negative. 52W high from fast_info. Days_below computed from price history.
13	Deal delay risk	Claude API web search	Searches latest earnings transcript for: "deal delay", "push out", "timing slippage", "deferred". Boolean flag → –1 if found, 0 otherwise.
14	Insider ownership	SEC EDGAR Form 4 (US) / SEDI (Canada)	>15% = founder-level alignment = +2. 1–15% = neutral = 0. <1% = treating it as a job = –1. Total insider ownership % from latest proxy filing.
15	Reinvestment	Claude API	Search: "[company] total addressable market TAM [year]".

5	runway	web search	Extracts management-stated TAM from earnings/investor day materials. TAM/revenue: $>10\times = +1$, $3-10\times = 0$, $<3\times = -1$. Source cited in stored rationale.
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TAM sourcing note: management-stated TAM from earnings calls and investor day materials is the primary source.

If not available in last 4 quarters, agent falls back to analyst report consensus. Source always cited in the stored rationale string.

SEC EDGAR: the US Securities and Exchange Commission public filing database. Free, no API key. Insider ownership from Form 4 filings.

For Canadian (TSX) stocks: SEDI (System for Electronic Disclosure by Insiders) is the Canadian equivalent of EDGAR for insider filings.

Divergence signal: the 60-day filter prevents fresh price breakdowns from triggering a false positive. A stock breaking down today is not a divergence setup.

5. Sector Calibration — Why Numbers Differ by Industry

The same financial ratio means completely different things in different industries. The engine uses different measuring sticks for each business type. Using universal thresholds creates systematic errors for non-SaaS businesses.

5.1 OEY Thresholds

business_type	OEY strong	OEY expensive	Rationale
saas	≥5%	<1%	Asset-light. Minimal capex. FCF ≈ true owner earnings.
compounder	≥4%	<1%	Moderate capex. Still predominantly cash-generative.
cyclical	≥4% (mid-cycle only)	<0.5%	Single-year OEY misleads at peak cycle.
income	≥3%	<1%	Heavy regulated capex. Lower FCF yield is structurally acceptable.

5.2 Net Debt/EBITDA Thresholds

business_type	Safe	Concern	Danger	Rationale
saas/compounder	Net cash (≤0×)	≤1.5×	≥5×	Asset-light, uncontracted cash flows
income	Net cash	≤2.5×	≥7×	Contracted cash flows service higher leverage
cyclical	Net cash	≤1.0×	≥3.5×	Commodity EBITDA can halve in a downturn

5.3 WACC (Weighted Average Cost of Capital)

business_type	WACC	Rationale
income	5.5%	Regulated returns, contracted cash flows, low risk premium
compounder	7.5%	Moderate risk, stable but uncontracted revenues
cyclical	9.0%	Commodity/cycle risk, earnings volatility
saas	10.0%	Rate-sensitive, far-future cash flows, high risk premium

5.4 EV/Revenue Valuation Tax Tiers

A pipeline at 4× EV/rev is structurally normal. A SaaS company at 4× EV/rev is moderately valued. A cyclical miner at 4× EV/rev is expensive. Different floor triggers per type.

business_type	Tier 1 (–1)	Tier 2 (–2)	Tier 3 (–3)	Tier 4 (–4)	Floor trigger (min –5)
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saas	≥3×	≥5×	≥8×	≥12×	≥4×
compounder	≥1.5×	≥3×	≥6×	≥8×	≥3×
income	≥4×	≥6×	≥8×	≥10×	≥6×
cyclical	≥1×	≥2.5×	≥4×	≥5×	None (no floor)

5.5 Rule of 40 and Margin Quality Substitution

business_type	margin_quality dimension uses	Rationale
saas	Rule of 40 (rev_growth% + FCF_margin%)	≥40 = top tier. Measures growth + profitability simultaneously.
compounder	Rule of 40	Applicable to growing cash-generative businesses.
income	Distribution coverage ratio (FCF / annual_div_obligation)	<1.0× = dividend is debt-funded. REPLACES Rule of 40 which gives pipelines a meaningless score.
cyclical	Normalised FCF margin vs cycle proxy	Checks if FCF margin is positive and revenue growing vs negative/contracting at trough.

6. Position Layers and Timeframe Architecture

6.1 The Core Problem This Solves

The quant model fires signals at different time horizons using different analytical functions. The key is that every signal carries a `signal_timeframe` parameter ('long' or 'short'), set by the function that fired it based on the interval it is running on. Long-interval signals act on the structural core position. Short-interval signals act on a separate tactical layer. They are tracked independently and close independently.

`signal_timeframe` rule: if the quant function is running on a short interval (daily/weekly candles), it is a short-interval signal acting on the tactical layer. If running on a monthly/quarterly/yearly interval, it is a long-interval signal acting on the core layer.

6.2 Four Conflict Scenarios

Scenario	Situation	Engine behaviour
Short SELL vs open long	Short-interval function (e.g. TrendPulse) fires 3-week overbought exit. DeltaDrift long still active.	Close tactical layer (take profit on \$2,500). Core \$7,500 untouched. Verdict: FULL EXIT or PARTIAL EXIT on tactical layer only.
Short BUY, long still open	Momentum bounce. Long-term thesis unchanged. Conviction still $\geq +5$.	Allow tactical add with flag "TACTICAL ADD to existing core position". Sum (core + tactical) capped at 1.0 of max position size.
Short BUY, conviction deteriorated	TrendPulse fires BUY. But since the core was opened, conviction has dropped to +1.	Engine flags TACTICAL_ADD_WARNING: conviction deteriorated. Tactical sizing automatically halved. Analyst should review core thesis.
Short BUY, long near trailing stop	Bounce signal fires. But quant model has flagged long-interval position near its trailing stop.	Engine fires TRAILING_STOP_WARNING and CANCELS the tactical add entirely. Do not add exposure just as the structural trade is about to exit.
Long SELL fires	Structural thesis has ended. Core position exits per conviction verdict.	Core exits per PAUSE/PARTIAL/FULL/HARD schedule. Tactical layer also closed — if structural case weakens, tactical must close too.

6.3 Position Layer Concrete Example — \$100,000 Portfolio

Maximum intended position in KXS.TO: 10% of portfolio = \$10,000.

Event	Action	core_frac tion	tactical_fra ction	\$ in KXS
DeltaDrift fires long-interval BUY. Conviction +6, <code>fd_direction=positive</code> → TACTICAL BUY 85%	Set core = 0.85	0.85	0.00	\$8,500
TrendPulse fires short-interval BUY. Conviction +6 → allow tactical add	Set tactical = 0.15 (cap: 1.0 - 0.85)	0.85	0.15	\$10,000 (fully sized)

TrendPulse fires short-term exit. Conviction still +6.	Zero tactical only. Core untouched.	0.85	0.00	\$8,500
DeltaDrift fires long-interval SELL. Conviction +6 → PARTIAL EXIT retain 50%	Core × 0.50. Tactical already 0.	0.43	0.00	\$4,300
Conviction drops to -3 (deteriorating fundamentals). Long SELL fires.	HARD EXIT. Core to zero. Tactical to zero.	0.00	0.00	\$0

6.4 Integration Pattern — Any Quant Model Function

Every quant model function passes its own interval to determine `signal_timeframe`. The conviction engine does not infer this — it receives it as a parameter.

```
# Any quant function fires → call modify_signal with interval context
# Short-interval function running on daily candles → short-term, tactical layer:
result = modify_signal('KXS.TO', 'BUY',
    signal_timeframe='short-interval',
    quant_model_name='BAND_MATRIX_1d_period20',
    signal_strength=0.75)

# Same function running on monthly candles → long-term, core layer:
result = modify_signal('KXS.TO', 'BUY',
    signal_timeframe='long-interval',
    quant_model_name='BAND_MATRIX_1mo_period20',
    signal_strength=0.80)

# If long position is near trailing stop, engine cancels tactical add:
result = modify_signal('KXS.TO', 'BUY', signal_timeframe='short-interval',
    quant_model_name='BAND_MATRIX_1d',
    long_position_near_stop=True)
```

7. Data Architecture — Overview

The engine persists data in a lightweight flat-file store — one JSON record per ticker, held in `conviction_store/{TICKER}.json`. Each record is divided into three zones:

- Static fundamentals (updated quarterly, after earnings): BQ scores and all sub-components, ROIC 5Y average, PE history array, EPS/FCF/net debt/forward revenue, `fd_direction` votes, business type, and all manual override values. These are computed once per quarter and read many times — keeping the daily runner fast.
- Dynamic daily slice (updated every morning before market open): current price, PE TTM, PE percentile vs 20Y history, EV/fwd revenue, Owner Earnings Yield, dividend yield and z-score, `conviction_score`, `fs_score`, `fs_class`, `yield_trap` flag, and any active alert flags.
- Position layer state (updated on each signal): `core_fraction`, `tactical_fraction`, signal dates, and the quant model name that opened each layer.

This split is deliberate. A full financials pull via `yfinance` takes 500ms–2s per ticker. Running that for 200 tickers daily would take 100–400 seconds pre-market. By caching fundamentals quarterly, the daily runner fetches only a price update per ticker — completing the full 200-ticker universe update in under 30 seconds.

At the current scale of 200 assets, the store is approximately 600KB of JSON — trivially small, human-readable, and version-controllable with no infrastructure required. The schema is designed so that swapping to a SQL database at larger scale (2,000+ assets) requires changing only the load/save functions — the data structure itself remains identical.

8. Complete Decision Tables

8.1 BUY-Side Verdict Mapping

conviction_score (after FS cap)	Verdict	Base sizing	fd=positive	fd=negative	Notes
≥+8	MAX CONVICTION	100%	100%	100%	yield_trap → CANCEL BUY (hard gate)
≥+5	TACTICAL BUY	75%	85%	60%	yield_trap → CANCEL BUY
≥+2	REDUCED BUY	40%	50%	25%	yield_trap → CANCEL BUY
<+2	CANCEL BUY	0%	0%	0%	yield_trap also → CANCEL BUY
Any (FS=weak, long-interval signal)	CANCEL BUY	0%	—	—	FS=weak (long): always CANCEL BUY regardless

8.2 SELL-Side Verdict Mapping

For long-interval signals: sizing_pct = fraction of CORE position to retain (0 = full exit). For short-interval signals: closes tactical layer only regardless of verdict name.

conviction_score	Verdict (long-interval signal)	Retain (core)	Short-interval signal behaviour
≥+8	PAUSE SELL	70%	Close tactical only. Core untouched.
≥+5	PARTIAL EXIT	50%	Close tactical only. Core untouched.
≥+2	FULL EXIT	0%	Close tactical. Core to 0%.
<+2	HARD EXIT	0%	Close both. Exit quickly.
Yield trap	HARD EXIT	0%	Hard gate — both layers closed.

8.3 FS Cap by Timeframe

fs_class	Long-interval signal — cap	Short-interval signal — cap	Rationale
strong	No cap	No cap	Full conviction applies
moderate_high	No cap	No cap	Full conviction applies
moderate	No cap	No cap	Normal — proceed
moderate_low	min(conviction, +4)	No cap	Structural hold needs quality bar; tactical trade less so

weak	min(conviction, +1)	min(conviction, +2)	Deteriorating business. Short: small tactical ok. Long: almost never buy.
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8.4 Asset Type Gate — First Check

Asset	quoteType	Engine output
Individual equity (stock)	EQUITY	Full BQ / FS / conviction calculation
Sector ETF (IGV, SPY etc)	ETF	NOT_APPLICABLE — ETF financials are fund-level aggregates, not operational
Index	INDEX	NOT_APPLICABLE
FX pair (EUR/USD)	CURRENCY	NOT_APPLICABLE — no fundamental modifier for currencies
Cryptocurrency	CRYPTOCURRENCY	NOT_APPLICABLE
Commodity future	FUTURE	NOT_APPLICABLE
Mutual fund	MUTUALFUND	NOT_APPLICABLE

9. TELUS Worked Example (T.TO)

TELUS is the reference example for the income business type. It demonstrates sector calibration working correctly — different thresholds, different dimension substitution, yield trap firing.

9.1 Key Financials (2024 Annual Report + Apr 2026)

Metric	Value	Context
Revenue	CAD \$20.4B	Mature Canadian telecom
EBITDA	CAD \$8.2B	Solid but capex-intensive
Net Income	CAD \$0.74B	Suppressed by D&A; on \$50B asset base
FCF	CAD \$1.6B	After heavy 5G/fibre capex spend
Total Debt	CAD \$27.5B	Funded 5G/fibre rollout
Net Debt / EBITDA	3.28×	Income safe $\leq 2.5\times$ — marginally elevated
Dividend Yield	7.3% (~\$1.61/share)	TSX market absolute threshold: 7%
FCF Dividend Coverage	0.87×	Dividend is 87% covered — gap funded by debt or asset sales
Payout Ratio	220% of net income	Remainder covered by D&A; cash + borrowing
ROIC (5Y avg)	5.5%	Exactly at income WACC 5.5% — zero value creation spread
EV / Fwd Revenue	~2.5×	Income tier 1 = 4× → no entry multiple penalty
Altman Z	~1.4	Below 1.8 = distress zone
Auto-detected type	income	Detected via: yield 7.3% > 2.5%, payout 220% > 45%, industry='telecom'

9.2 BQ Score Breakdown

Dimension	Score	Notes (income sector — calibrated thresholds)
revenue_quality	+1	Gross 39% ok for telecom; FCF conv 2.16× (D&A-driven); retention 88% < 95%
growth_trajectory	0	Revenue +3.2% — mature saturated Canadian telecom market
margin_quality (INCOME TYPE: distribution coverage)	-1	FCF coverage 0.87× < 1.0× = dividend partially debt-funded. Rule of 40 NOT applied (income type substitution)
balance_sheet	-1	Net debt/EBITDA 3.28× (income safe $\leq 2.5\times$ = neutral). Altman Z 1.4 < 1.8 = -1 penalty
roic_wacc_spread	0	ROIC 5.5% vs income WACC 5.5% = 0% spread. Exactly at cost of capital — not creating value
gross_margin_trend	-1	Telecom margins structurally compressing — more competition, fibre opex

debt_maturity_risk	0	Long-dated bonds — near-term refinancing risk manageable despite high leverage
ceo_quality	-1	Score 4/10 + new CEO transition penalty. Entwistle retired after 24 years.
mgmt_capital_allocation	-1	Score 3/10 — aggressive leverage build; 5G ROI timeline uncertain
competitive_moat	0	Score 4/10 — regulated duopoly but weak vs Rogers on customer loyalty
macro_tailwind	-1	Rate sensitivity: \$27B debt. Every 100bp rate move = ~\$270M incremental interest cost
divergence_signal	0	Price fell AND fundamentals weakened simultaneously. Not a divergence setup.
deal_delay_risk	0	N/A for telecom subscription
insider_ownership	0	~1% management ownership — treating it as a job, not an investment
reinvestment_runway	0	3× TAM/rev (mature market) → scores 0 per calibration
BQ_RAW TOTAL	-5	

9.3 Valuation Tax and Conviction

Component	Score	Notes
entry_multiple (income tiers: 4/6/8/10×)	0	EV/rev 2.5× below tier 1 (4×) — no penalty for income
pe_hist_percentile	-2	PE 34× at ~82nd pctile of own history — near high end
growth_multiple_fragility	0	Revenue growth 3.2% — not fragile at this multiple
business_type relief (+1 for cheap income)	+1	EV/rev below tier 1 → income gets modest relief
deal_delay_signal / market_regime_beta	0	N/A
VALUATION TAX TOTAL	-1	No -5 floor — income floor trigger is 6×; EV/rev is 2.5×

BQ_RAW	-5	
Valuation tax	-1	
conviction_raw	-6	
fs_quality_base	37.5	50 + (-5 × 2.5)
fs_score	~38.5	After OEY 6.4% (income strong ≥3% → +5) and PE pctile adjustments
fs_class	moderate_low	Score 38.5 → moderate_low (between 25 and 40)

FS cap applied?	No cap needed	conviction -6 is already below the +4 cap
FINAL conviction_score	-6	HARD EXIT if holding. CANCEL BUY on any signal.

9.4 Yield Trap Check

Condition	Value	Threshold	Fires?
Z-score (yield vs 5Y own history)	2.00	Must be >1.5	YES
Market absolute threshold (TSX)	7.3% yield	7% (TSX)	YES — marginally above
FCF dividend coverage	0.87×	Must be ≥1.2×	YES
YIELD TRAP RESULT	FIRES — hard gate engaged		

YIELD TRAP FIRES — hard gate engaged

- BUY signal → CANCEL BUY (yield trap hard gate overrides conviction score entirely)
- SELL signal → HARD EXIT (yield trap fires on sell side too)

Note on OEY: OEY 6.4% looks attractive on the surface. But FCF is suppressed by a capex cycle expected to last 3–5 more years. Normalised OEY is closer to 4–5%. More importantly: 13% of the dividend is debt-funded — the real sustainable yield on today's business economics is lower than 7.3%.

10. Operational Guide

10.1 Morning Cron — Pre-Market Sequence

Step	Function	What it does	Time
1	check_earnings_triggers()	Checks earnings calendar. Fires <code>full_recalculation()</code> for any tickers whose earnings date has passed.	2–5 min per trigger
2	run_daily_universe(manifest)	Calls <code>daily_update()</code> for all 200+ assets. ONE price-only API call per ticker. Returns <code>alert_map {ticker:[flags]}</code> .	~20 seconds
3	generate_daily_report(alert_map)	Formats Fundamental Alerts for Claude daily briefing. Universe health + conviction distribution + flagged tickers.	<1 second

```
Crontab (6am Mon–Fri): 0 6 * * 1-5 /usr/bin/python3 /path/to/conviction_engine_v5_final.py --cron
```

10.2 Post-Earnings Checklist — What to Update After Each Call

Event	Function to call	Key fields to update
Earnings call reviewed	<code>update_overrides(ticker, {...})</code>	<code>deal_delay_flag</code> , <code>revenue_accelerating</code> , <code>divergence_signal</code> , <code>retention_pct</code>
CEO/CFO changed	<code>update_overrides(ticker, {...})</code>	<code>new_ceo_transition=True</code> , <code>ceo_quality_score=new_value</code> , <code>mgmt_alloc_score=new_value</code>
New earnings date announced	<code>set_earnings_date(ticker, 'YYYY-MM-DD')</code>	Cron auto-fires <code>full_recalculation()</code> when date passes
PE history thin (<20 pts, US ticker)	<code>fetch_pe_history_macrorends(ticker, slug)</code>	Auto-called in <code>full_recalculation</code> if <20 pts and US ticker
Non-US PE history needed	Edit store JSON directly	Set <code>pe_20y_array</code> via Gurufocus / TIKR / Screener.in / manual
New asset added to universe	<code>full_recalculation(ticker, overrides)</code>	All fields populated. Run once then daily runner keeps warm.

10.2 Signal Timeframe — Definitive Function Mapping

The table below shows which interval each function operates on, and therefore which position layer its signals act on. Short-interval functions (daily/weekly) act on the tactical layer. Long-interval functions (monthly/quarterly/yearly) act on the core layer.

■ IMPORTANT — SHORT-INTERVAL vs LONG-INTERVAL SIGNALS IN THIS ENGINE:

When this document says short-interval signal or long-interval signal, this refers to a signal fired by a quant function

running on a SHORT interval (e.g. interval=daily — signals play out over days to weeks) versus a LONG interval (e.g. interval=monthly — signals play out over months). The signal_timeframe parameter in modify_signal() is set by the quant function based on its own interval.

The conviction engine does NOT determine the timeframe — the quant function determines it and passes it in.

Short-interval → tactical layer (added on top of core, closes independently)

Long-interval → core layer (structural position, held for months to years)

10.2 Signal Timeframe — Definitive Function Mapping

The table below shows which timeframe each quant function operates on. Short-interval functions act on the tactical layer; long-interval functions act on the core layer. The conviction engine receives signal_timeframe as a parameter — it does not determine it.

Function	Source file	Interval (from code)	signal_timeframe
OSCILLATOR_DELTA	divergence_comp.py	INTERVAL_DAILY (fixed)	'short'
TRENDPULSE	trendline.py	All intervals — D/W/M/Q/Y	'short-interval' if D/W; 'long-interval' if M/Q/Y
BASELINEDIVERGENCE	general_divergence.py	INTERVAL_WEEKLY (fixed)	'short'
FRACTAL_TRACK	fib_ret.py	D/W/M/Q (selectable)	'short-interval' if D/W; 'long-interval' if M/Q
BAND_MATRIX	bollinger.py	All intervals (selectable)	'short-interval' if D/W; 'long-interval' if M/Q/Y
DELTADRIFT	distance.py	All intervals (selectable)	'short-interval' if D/W; 'long-interval' if M/Q/Y
SIGMASHELL	sigma.py	INTERVAL_DAILY (fixed)	'short'
PULSEGAUGE	sentiment.py	INTERVAL_WEEKLY (fixed)	'short'
ALTITUDE_ALPHA	new_high.py	All intervals (selectable)	'short-interval' if D/W; 'long-interval' if M/Q/Y
COMBINED_STRATEGY	combined_strategy.py	All intervals (selectable)	Set per underlying sub-function interval

10.3 Integration Point — trade_arrival_analysis.py is the dispatcher

trade_arrival_analysis.py already acts as the signal orchestrator: it runs all registered functions and collects their signals into signallist[]. This is the correct integration point — one location, not 10 separate files.

Step 1: Create conviction_dispatch.py in the project root

```

from conviction_engine_v5_final import modify_signal, format_signal_table

LONG_INTERVALS = {'Monthly', 'Quarterly', 'Yearly', '1mo', '3mo', '1q', '1y'}

def conviction_dispatch(symbol, signal_value, interval, function_name,
                       strength=0.75, near_stop=False):
    if signal_value == 0: return None
    direction = 'BUY' if signal_value == 1 else 'SELL'
    timeframe = 'long-interval' if interval in LONG_INTERVALS else 'short-interval'
    try:
        result = modify_signal(ticker=symbol, technical_signal=direction,
                               signal_timeframe=timeframe, signal_strength=strength,
                               quant_model_name=f'{function_name}_{interval}',
                               long_position_near_stop=near_stop)
        print(format_signal_table(symbol, direction,
                                  quant_model=f'{function_name}_{interval}',
                                  signal_timeframe=timeframe, signal_strength=strength))
    except RuntimeError as e:
        print("[conviction] " + e + " — run full_recalculation first")
    return result

```

Step 2: In trade_arrival_analysis.py — add import + single call

```

# At top of trade_arrival_analysis.py:
from conviction_dispatch import conviction_dispatch

# In the signal collection loop, immediately after signallist.append(signal):
conviction_dispatch(
    symbol      = symbol,
    signal_value = signal,      # already 1 or -1 at this point
    interval    = config.get("interval") or interval,
    function_name = function_name, # key from FUNCTION_CONFIG
    strength    = win_rate,      # function own win_rate as signal strength
)

# That is the complete integration. Two steps: one new file, one added call.
# All 10 registered functions route through this automatically.

```

10.4 CLI Quick Reference

Command	What it does
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python conviction_engine_v5_final.py KXS.TO BUY	Snapshot + signal table for KXS BUY
python conviction_engine_v5_final.py -- cron	Full morning sequence
python conviction_engine_v5_final.py -- report	Daily report only
python conviction_engine_v5_final.py -- init universe.txt	Day 0 initialisation for all tickers
python conviction_engine_v5_final.py -- set-earnings KXS.TO 2026-07-15	Set earnings date
python conviction_engine_v5_final.py -- overrides KXS.TO	Print current manual overrides
python telus_test.py	Run TELUS mock test (no network needed)

— End of document —